

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)
CHARUSAT Campus, Changa – 388 421.

M Sc (BIOCHEMISTRY)
SEMESTER III EXAMINATION (BACKLOG)
BC802: CLINICAL BIOCHEMISTRY

Date: 22.04.2011

Time: 10.00 AM to 1.00 PM

Total Marks: 70

Instructions to the candidates:

1. Provide all necessary information as required in the answer sheet.
2. Attempt all questions.
3. Questions in **Part A** should be answered in the question paper itself. Please **do not write** your candidate ID number on the question paper for Part A.
4. Write answers for Section I and Section II of **Part B** in separate answer sheets.
5. Use of cell phones is strictly prohibited.
6. Use of non-programmable calculators may be allowed if needed.

PART A

Total marks: 20

Q1 (A) Choose the correct option and put ✓ mark in front of it:

1 Frothy urine is characteristic of

- a. proteinuria
- b. ascites
- c. edema
- d. none of the above

2 Inability of the kidneys to get rid of nitrogenous wastes leading to their accumulation in the body is known as

- a. Hyperkalaemia
- b. Azotemia
- c. Glomerulonephritis
- d. rickets

3 Interruption or stagnation of the bile flow in any part of the biliary system is known as

- a. Hepatitis
- b. Wilson's Disease
- c. Cholestasis
- d. Steatohepatitis

4 Any one of the following causes inflammation of the liver resulting from infection with a DNA-type virus.

- a. Hepatitis A
- b. Hepatitis B
- c. Hepatitis C
- d. Hepatitis E

5 An infant had following physical features: up-slanting palpebral fissures, bilateral epicanthal folds, flat nasal bridge, open mouth with tendency of tongue protrusion, and small ear with overfolded helix. Probable diagnosis of this patient is

- a. Down Syndrome
- b. Cri-du chat Syndrome
- c. Patau Syndrome
- d. Edward Syndrome

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6 Microcytic anaemia is the result of deficiency of any one of the following

- a. Vitamin B12
- b. Iron
- c. Folic acid
- d. Calcium

7 The loss of all four alpha genes of hemoglobin produces a condition that is incompatible with life. The gamma chains produced during fetal life associate in groups of four to form an abnormal hemoglobin called "hemoglobin Barts", characteristic of one of the following diseases:

- a. Megaloblastic anaemia
- b. Microcytic anaemia
- c. Alpha Thalassemia
- d. Beta Thalassemia

8 Acanthosis nigricans meaning brown to black, velvety hyperpigmentation of the skin is a symptom in one of the following endocrine disorders

- a. Cushing's syndrome
- b. Acromegaly
- c. Grave's disease
- d. Hashimoto's thyroiditis

9 This carbohydrate deficiency is associated with cataracts

- a. GALK deficiency
- b. GALT Deficiency
- c. Hyperglycemia
- d. None of the above

10 One of the following is a hallmark of any urea cycle disorders:

- a. $>$ ammonia
- b. $<$ ammonia
- c. $>$ carbon dioxide
- d. None of the above

11 Ochronosis is a characteristic of any one of the following disorder

- a. Alkaptonuria
- b. N-acetyl glutamate synthetase deficiency
- c. Starvation
- d. Obesity

12 Albinism is caused because of error in the metabolism of one of the following amino acid

- a. Methionine
- b. Tryptophan
- c. Valine
- d. Tyrosine

13 Familial hypercholesterolemia occurs because of mutation in

- a. HDL receptor gene
- b. LDL receptor gene
- c. F8 gene
- d. None of the above

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Q1 (B) State True or False

(4)

1. Degradation errors in any of the glycogen storage disorders may lead to hypoglycaemia
2. Hyperepinephrinemia is one of the complication in persistent hypoglycemia
3. Majority of Noninsulin-dependent diabetes mellitus is attributed to obesity
4. Accumulation of propionic acid in Propionic Acidemia leads to accumulation of free ammonia

Q1 (C) Complete the following table:

(3)

Name of the disease	Deficient enzyme
Fabry's disease	
	Ceramidase
Sandhoff's disease	

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Part B

Total Marks: 50

Section I

Q1 (A) Name two enzymes involved in the formation of bilirubin. Explain steps in formation of conjugated bilirubin (4)

Q1. (B) Write short note on any two: (4)

- a. Steatosis
- b. Wilson's disease
- c. Gilbert's disease

Q1. (C) Write short note on: Alcoholic liver disease (2)

OR

Q1. (C) Jaundice (2)

Q2. (A) Clinical findings in HYPERCHYLOMICRONEMIA (4)

OR

Q2. (A) HYPERTRIGLYCERIDEMIA

Q2. (B) Tay-Sach disease (4)

OR

Q2. (B) Metachromatic Leukodystrophy (4)

Q2. (C) Niemann-Pick (2)

OR

Q2. (C) Gaucher's disease (2)

Section II

Q3. (A) G6PD deficiency

(4)

OR

Q3. (A) Complete the following table with respect to glycogen storage diseases

(4)

Type	Affected Organ	Deficient enzyme
Type VI, Hers disease		
Type V, McArdle's disease		
		Amylo-1,4 1,6-transglucosidase [Glycogen branching enzyme]
		Amylo- 1,6-glucosidase [Glycogen debranching enzyme]

Q3. (B) Pathological mechanisms in development of diabetic retinopathy

(4)

OR

Q3. (B) Macrovascular Complications in Diabetes

(4)

Q3. (C) Aldolase B deficiency

(2)

OR

Q3. (C) Classic Galactosemia

(2)

Q4. (A) Cushing's syndrome

(4)

OR

Q4. (A) Biochemical basis of megaloblastic anaemia

(4)

Q4. (B) Physical abnormalities in Turner's syndrome

(4)

Q4. (C) Answer any one:

(2)

1. Edward's syndrome
2. Molecular basis of sickle cell anaemia

Q5. (A) Complete the following sequence of reaction writing enzyme and its corresponding disorder:

(4)

Phenylalanine -----enzyme 1/disorder 1-----> Tyrosine-----enzyme 2/disorder 2-----> p-
OH phenylpyruvate -----enzyme 3/disorder3----->homogentisate -----enzyme 4/disorder 4 -
----> maleylacetoacetate

Q5. (B) Write on any 2

(4)

1. MSUD
2. Homocystinuria
3. Marasmus

Q5. (C) Write on any one:

(2)

1. HGPRT deficiency
2. Orotic aciduria